Available online at www.sciencedirect.com

ScienceDirect

Advances in Space Research 75 (2025) 4588–4599

ADVANCES IN SPACE RESEARCH
(a COSPAR publication)

www.elsevier.com/locate/asr

Spatio-temporal evaluation of tropospheric delay products in China using CMONOC data as reference

Zhuoyue Wen [a], Fei Yang [a,*], Jian Li [b], Ruixian Hao [a], Tianyang Liu [a], Zhuangzhuang Wang [a]

[a] *College of Geoscience and Surveying Engineering, China University of Mining and Technology-Beijing, Beijing 100083 China*
[b] *School of Electronic and Information Engineering, Beihang University, Beijing 100191 China*

Received 1 November 2024; received in revised form 5 December 2024; accepted 28 December 2024
Available online 2 January 2025

Abstract

The delay of the neutral atmosphere to the Global Navigation Satellite System (GNSS) signal, often expressed as the zenith tropospheric delay (ZTD), is critical to the accuracy and convergence time of GNSS positioning and is the basis for precipitable water inversion in GNSS meteorology. By projecting the tropospheric delay to the zenith direction through a mapping function, various ZTD models, as well as ZTD products, have been proposed and applied to compensate for this error. In this study, two tropospheric products, TUW-VMF3 and GFZ-VMF3, developed by the Vienna University of Technology (TU Wien) and the GeoForschungsZentrum Potsdam (GFZ), are evaluated temporally and spatially in China. The ZTD data used in this study for comparison are from the Crustal Movement Observation Network of China (CMONOC) covering 238 GNSS stations for the period from 2012 to 2020. The analysis considered the impact of geographical distribution, season, and epochs on the performance of the products and examined their performance across different climatic regions in China. Additionally, the study explored the performance of products with different spatial resolutions (1.0°×1.0° and 5.0°×5.0°) in estimating ZTD, revealing their potential advantages and limitations in practical GNSS applications. The results revealed that the two products at the same spatial resolution exhibited high similarity with Root Mean Square Error (RMSE) and bias of 17.90 /1.82 mm and 18.87 /1.42 mm for the TUW-VMF3 and GFZ-VMF3, respectively. The TUW-VMF3 outperformed the GFZ-VMF3 at 89.1 % of the stations, but noted that the RMSE difference between the two was very small, especially in the temperate continental (TC) and temperate monsoon (TM) regions. Considering that GFZ-VMF3 held an advantage on 10.9 % of the stations, particularly in the northwest regions of SM and TC, so it was advised that the selection of products for practical application should be based on regional performance. Compared to the TUW-VMF3-5.0°×5.0°, the TUW-VMF3-1.0°×1.0° demonstrated a significant improvement in accuracy at 93.3 % of the stations, and performed better in the subtropical monsoon (SM) and TC regions, but was outperformed by the TUW-VMF3-5.0°×5.0° in certain stations located in extreme high-altitude or extremely cold regions. These results indicate that employing ZTD products with higher spatial resolution can significantly enhance the accuracy of ZTD forecasting in GNSS applications. It is noteworthy that all products performed poorest in the SM region, especially in summer. The difference in bias values between the products was not significant. Furthermore, the study found that the RMSE values of all products decreased with increasing altitude and latitude, with the most pronounced change observed in the GFZ-VMF3. Time series analysis of ZTD at stations with extreme RMSE values showed that ZTD values peaked in the summer and were at their lowest in winter.

Keywords: GNSS; Zenith tropospheric delay; Tropospheric delay products; ZTD dataset

* Corresponding author at: No. 11, Xueyuan Road, Haidian District, Beijing, China.
E-mail address: yangfei@cumtb.edu.cn (F. Yang).

https://doi.org/10.1016/j.asr.2024.12.070
0273-1177/

1. Introduction

The Global Navigation Satellite System (GNSS) signal is delayed as it travels from the satellite through the neutral atmosphere to the receiver, known as the tropospheric delay (Bevis et al., 1992; Yang et al., 2021a; Yao et al., 2018). It is one of the key factors affecting the performance of high-precision satellite navigation, positioning, and water vapor inversion (Wang et al., 2022; Penna et al., 2001). The delay varies from 2 m to 20 m due to differences in satellite elevation angles (Chen et al., 2020; Penna et al., 2001) and exhibits changes related to latitude and altitude (He et al., 2024). It is common to use an elevation-dependent mapping function to the zenith direction (Yang et al., 2023a), with the zenith tropospheric delay (ZTD) used to characterize the effect of the troposphere on signal propagation. It has been shown that an accurate ZTD has a crucial impact on GNSS position and navigation in terms of convergence time and accuracy (Zumberge et al., 1997). Therefore, it is particularly important to establish a model that provides a stable and accurate ZTD.

Several models have been developed over the years for the quantification and mitigation of tropospheric delays (Jiang et al., 2020). Tropospheric delay models such as the Hopfield model, the Saastamoinen model, and the Black model are capable of obtaining ZTD values with cm-level accuracy through feeding in accurate measured meteorological observations and positional parameters (Black, 1978; Hopfield, 1969; Saastamoinen, 1972). However, many GNSS stations are not equipped with meteorological sensors or are unable to provide accurate meteorological parameters. Therefore, researchers have developed various tropospheric delay models that do not require meteorological parameters to enhance positioning accuracy. One category is empirical models based on reanalysis meteorological data. For instance, Li et al. (2012) established the IGGtrop series models by using the reanalysis data derived from the National Centers for Environmental Prediction (NCEP). Based on the European Center for Medium-Range Weather Forecasts (ECMWF) reanalysis data, Song et al. (2011) constructed the regional tropospheric grid model SHAO-C for the Chinese region, Yao et al. (2013) further developed the Global Zenith Tropospheric Delay (GZTD) model. But the reanalysis meteorological datasets are plagued by data upload delays, which may not meet the real-time requirements of meteorological research (Yang et al., 2024). Another category of models is based on the measured GNSS ZTD data, for example, Chen et al. (2019) established the Shanghai Astronomical Observatory Troposphere Model (SHATrop) for China based on the ZTD products from 223 stations of the Crustal Movement Observation Network of China (CMONOC), He et al. (2024) developed a Global Tropospheric Delay forecasting model named LSTM-ZTD, which utilizes Long Short-Term Memory (LSTM) neural networks. The progression of these models signifies a substantial advancement within the domain of ZTD models, concurrently heralding a noteworthy enhancement in the precision and dependability of ZTD forecasting. Additionally, Numerical weather prediction models (NWMs) rely on physical algorithms that utilize fundamental parameters such as temperature, pressure, and water vapor pressure to estimate tropospheric delays. The UNB series models (Leandro et al., 2006; Leandro et al., 2008), GPT series models (Lagler et al., 2013; Böhm et al., 2015) and the EGNOS model (Penna et al., 2001) are typical representatives of this category. Additionally, the ongoing development and refinement of these models by numerous researchers have yielded significant outcomes (Yang et al., 2021b; Yang et al., 2021c), contributing substantially to the enhancement of their accuracy and applicability.

With the increasing demand for ZTD estimation, new ZTD products that combine physical methods from NWMs and actual GNSS data are emerging continuously, which further promote the application and development of high-precision delay models. Böhm et al. (2009) developed a VMF1 product (TUW-VMF1) based on ECMWF data, with a horizontal resolution of 1.0°×1.0°. The VMF3 product (TUW-VMF3), an enhanced iteration of VMF1, utilizes an improved ray tracer and achieving a higher horizontal resolution, which further improves the accuracy of ZTD estimation. The GeoForschungsZentrum Potsdam (GFZ) has developed a model analogous to VMF3, known as GFZ-VMF3, utilizing the fifth-generation ECMWF reanalysis (ERA5) data. This model matches the topography of the TUW-VMF3 with a horizontal resolution of 1.0°×1.0° (Zus et al., 2015). Both the TUW-VMF3 and GFZ-VMF3 products are highly regarded in various real-world applications due to their accuracy and reliability, and are widely used in PRIDE-PPP from Wuhan University, raPPPid, PPPH, PPP-ARISEN and MG-APP and other kinds of open-source software or platform (Geng et al.,2019; Glaner and Weber, 2023; Yao et al., 2024). In addition, Osah et al. (2021) employed the TUW-VMF3 as an alternative source for ZTD data, enhancing the positioning and meteorological applications of the International GNSS Service (IGS) final ZTD products in West Africa. Zhang et al. (2021) utilized the TUW-VMF3 and GFZ-VMF3 to estimate and correct the ZTD of GPS and GLONASS signals, thereby improving the precision of Precise Point Positioning (PPP). They also applied the MAZHA method to TUW-VMF3, further enhancing the accuracy of GPS-PWV inversion over the complex terrain of the Tibetan Plateau.

With the wide use of ZTD products, the analysis of their accuracy and availability has been conducted. For example, Yang et al. (2021) utilized IGS-ZTD products from approximately 400 globally distributed monitoring stations to assess the accuracy of the TUW-VMF3 product and the Single Point Positioning (SPP) biases caused by the residual ZTD (dZTD) after correction. Ssenyunzi et al. (2023) utilized six statistical assessment parameters to evaluate the performance of the TUW-VMF3 product in the African tropical region. Yao et al. (2024) assessed the global accuracy of the TUW-VMF3 product using the ZTD products provided by 495 IGS stations as reference values. Considering that previous research only analyzed the TUW-VMF3 products, Li et al. (2024) compared the accuracy of TUW-VMF3 and GRZ-VMF3 products in detail, but only the regions with dense GNSS stations, including North America, Europe, Japan, South America and Australia, were selected to conduct the comparisons. Although numerous scholars have conducted research on GNSS stations in the Chinese region in recent years, there has been relatively little analysis of ZTD products (Yang et al., 2023b; Wang et al., 2024). In this study, the performance of the two ZTD products in China was focused for the first time, with the aim of enhancing the accuracy of ZTD estimation for GNSS applications. Noted that the China spans a large latitude, with large differences in altitude and complex topography, these factors led to significant differences in the climatic conditions of different regions. Therefore, it focused on the unique geographical and climatic characteristics of China, which were often overlooked in global research, and the ZTD products with different spatial resolutions were considered in this experiment to better show their performance in different climate regions.

The paper is organized as follows: Section II describes the data and methods, Section III presents the analysis and discussion of the ZTD from TUW-VMF3/GFZ-VMF3 products, including the discussion of ZTD products with different spatial resolutions. Section IV describes the conclusions.

2. Data and methods

2.1. The ZTD products

The VMF service provides the VMF3 tropospheric product, which is a publicly available dataset generated using ray-tracing techniques. Two spatial resolutions, namely 1.0°×1.0° and 5.0°×5.0°, can be accessed at https://vmf.geo.tuwien.ac.at/trop_products/GRID/1×1/VMF3/VMF3_OP/ and the other at https://vmf.geo.tuwien.ac.at/trop_products/GRID/5×5/VMF3/VMF3_OP/, respectively.

The GFZ service develops the GFZ-VMF3 using an improved ray tracer and horizontal resolution, which is a new version of GFZ-VMF1 product (Zus et al., 2015). The GFZ-VMF3 data are available at ftp://ftp.gfz-potsdam.de/pub/home/GNSS/products/gfz-vmf1/.

Note that the two ZTD products have different temporal resolutions, GFZ-VMF3 provides ZTD at UTC 00:00, 03:00, 06:00, 09:00, 12:00, 15:00, 18:00 and 21:00 each day, and the ZTD products of TUW-VMF3 are provided only at UTC 00:00, 06:00, 12:00 and 18:00. To ensure the harmonization of time references, this study adopted the interpolation method referenced Li et al. (2024) to obtain hourly ZTD values covering 24 h. Linear interpolation, a straightforward and efficient technique, estimated the value at an intermediate point by assuming a uniform change between two known data points. It ensured that the ZTD values of the three datasets were consistent with the reference ZTD values across the 24 epochs, establishing a unified time frame for subsequent analyses and comparisons. Therefore, three types of ZTD products were utilized in this paper, which are listed in Table 1. In the table, the GFZ-VMF3 product was included, which provides ZTD estimates at 3-hour intervals throughout the day. The TUW-VMF3 was selected in two spatial resolutions (1.0°×1.0° and 5.0°×5.0°) to better analyze the impact of spatial resolution on the precision of ZTD estimations.

2.2. ZTD data from CMONOC

In this paper, the three ZTD products mentioned above are evaluated using China Crustal Motion Observation Network (CMONOC) as the reference. The CMONOC consists of more than 260 continuous stations covering the Chinese mainland, which allow continuous monitoring of ZTD over China as accurately as possible. The main purpose of CMONOC is to monitor crustal motions, gravitational field morphology and variations, and ground-based GNSS atmospheric water vapor monitoring over the Chinese mainland. Hourly ZTDs are calculated using observations from the CMONOC GNSS station and can be found at ftp://ftp.cgps.ac.cn/. In this study, 238 stations with data exceeding 60 % were selected for the nine-year time span from 2012 to 2020 based on the principles of low station data missing rate, and the coordinate time series has a large span and high stability. The geographical distribution of the stations is shown in Fig. 1.

2.3. Evaluation metrics

To more accurately compare the accuracy of the three ZTD products, Bias and Root Mean Square Error (RMSE) are used as evaluation metrics. Bias reflects the degree of deviation between the estimated and true values of the product, and a low bias indicates that the model predictions are close to the actual values. RMSE reflects the stability of the product, and a low RMSE indicates small prediction errors. The calculation formula is as follows:

$$Bias(\mathrm{mm}) = \frac{1}{n}\sum_{i=1}^{n}(ZTD_i - ZTD_i^0) \tag{1}$$

Table 1
The tropospheric correlation data used in this study.

Data	Provider	Spatial Resolution of model
GFZ-VMF3	GFZ	1.0°×1.0°
TUW-VMF3-1 × 1	TU Wien	1.0°×1.0°
TUW-VMF3-5 × 5	TU Wien	5.0°×5.0°

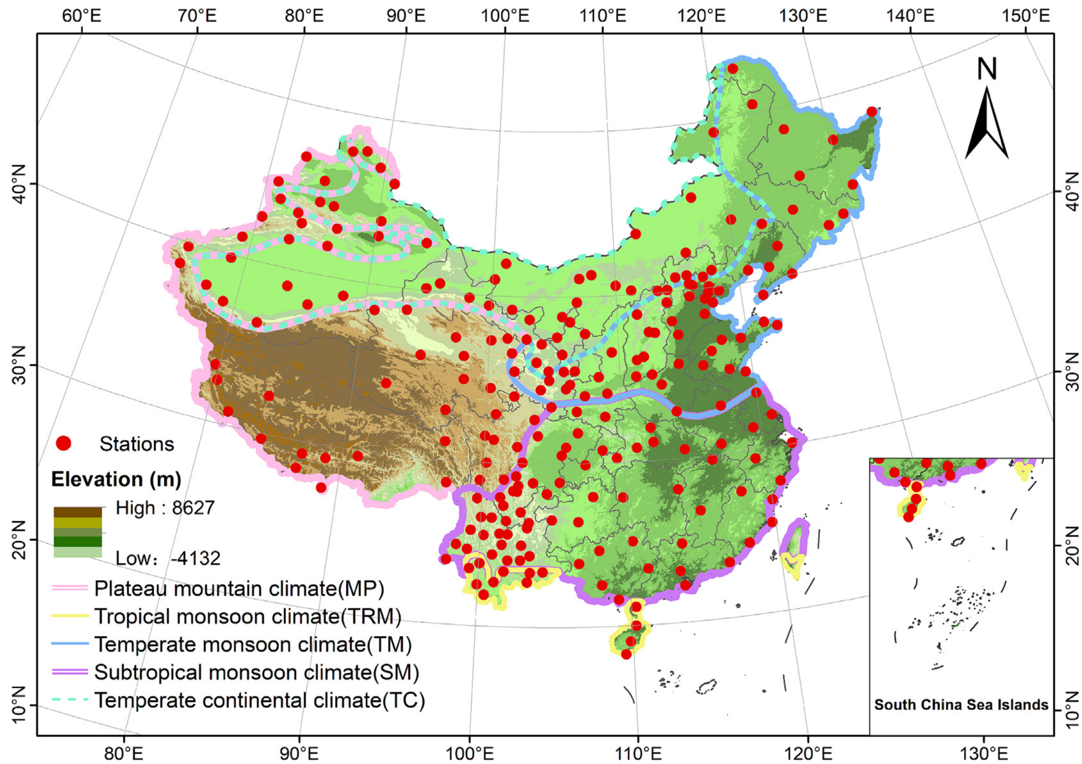

Fig. 1. Geographic distribution of the 238 GNSS stations used in this study.

$$RMSE(mm) = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(ZTD_i - ZTD_i^0)^2} \tag{2}$$

where ZTD_i is the estimated value of the product, ZTD_i^0 is the ZTD data provided by the CMONOC, and n is the cumulative number of hours.

3. Results and discussions

3.1. Overall analysis in China

The Fig. 2 displays a linear regression of the reference ZTD against the ZTD values of the three products, in which the colored shaded area indicates the range of the scatter distribution, and the black and red lines represent the fitted curve and the 1:1 line, respectively. The fitted curve represents the linear relationship between the ZTD predicted by the products and the reference ZTD. It is evident that the scatter plots of the ZTD values of three products are mostly concentrated around the 1:1 line. The difference between the GFZ-VMF3 and TUW-VMF3 in RMSE and bias is not significant, the values are 18.87/1.42 mm and 17.90/1.82 mm, respectively. When comparing products with different spatial resolutions, it can be seen that the accuracy of the TUW-VMF3-1.0°×1.0° is significantly better than that of the TUW-VMF3-5.0°×5.0°, especially with a 31.8 % improvement in RMSE. The improvement in ZTD accuracy of this magnitude can

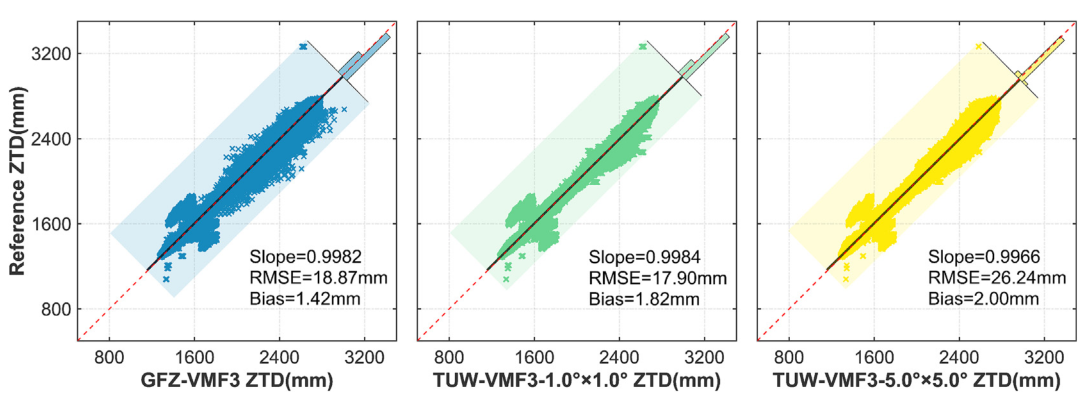

Fig. 2. Scatter plots of the three ZTD products and the reference values.

obviously enhance the accuracy of the conversion of ZTD to PWV in GNSS meteorology, and provide better tropospheric initial values for GNSS positioning.

The histogram in Fig. 3 shows the distribution of the differences between the three ZTD products and the referenced ZTD values from CMONOC. The σ is the standard deviation of the normal distribution, which describes the degree of dispersion of the data. The dashed line here indicates the mean of the normal distribution, reflecting the centralized trend of the data. It can be seen that all three products have negative mean values and their ZTD residuals all conform well to the normal distribution, with most of the residuals concentrated around the mean and decreasing towards the sides.The σ values of GFZ-VMF3 and TUW-VMF3 are 18.82 mm and 17.81 mm, respectively, and the percentages of residuals in the range of -50 mm to 50 mm are 98.4 % and 98.8 %, respectively. This indicates that the residuals of the two ZTD products with $1.0°\times1.0°$ are small and concentrated. Then comparing the products with different spatial resolutions, it can be seen that the residual distribution of TUW-VMF3-$1.0°\times1.0°$ is more concentrated than that of TUW-VMF3-$5.0°\times5.0°$. The percentage of residuals in the range of -15 mm $\sim$ 15 mm is 75.8 % instead of 59.5 %, which is 16.3 % higher, while σ is 31.9 % lower.

3.2. Analysis of different climate zones

The Fig. 4 shows the distribution of RMSE and bias for the three products in China, in which the stations are categorized into five research regions: plateau mountain (MP), tropical monsoon (TRM), temperate monsoon (TM), subtropical monsoon (SM), and temperate continental (TC). The number of stations in each region is 49, 13, 57, 74, and 45, respectively. Among them, the number of stations in the TM and SM regions are high and concentrated, accounting for 24.0 % and 31.1 % of the total stations, respectively. In contrast, the number of stations in the TRM region is lower, accounting for only 5.5 % of the total. For GFZ-VMF3 and TUW-VMF3, the majority of stations do not exceed 20 mm in terms of RMSE. When further analyzing the RMSE in each region, nearly 80 % of the stations in the TC and TM regions are below 15 mm, with more stations in TC below 10 mm. For the MP, SM, and TRM regions, over 20 mm is relatively more frequent but mostly between 10 mm and 20 mm. Comparing the RMSE of the different resolution products, the TUW-VMF3-$1.0°\times1.0°$ has a lot more stations within 15 mm than the TUW-VMF3-$5.0°\times5.0°$. Especially in the three regions of TM,TC,SM, the stations are improved to within 15 mm in 77.2 %, 73.3 %, and 57.3 %, respectively, reflecting the higher accuracy. In terms of bias, there is not much difference between the three products, with most of the values ranging from -10 mm to 10 mm, and more differences between regions. In the TM, TC, and SM regions, about 70 % of the stations show positive bias, and the TM region is more concentrated between 0 mm and 10 mm. While more negative bias is exhibited in the MP region, the probabilities of positive and negative bias are roughly equal in the TRM. These indicate that the three products overestimate the ZTD values in the TM, TC, and SM regions but underestimate the ZTD values in the MP region.

Fig. 5 compares the RMSE values of the three products in different climatic regions through the differences in color and arc length, which visually reflects their good or bad performance. Table.2 further lists the specific RMSE values in each region as references, which demonstrates the gaps between the three products in detail. For GFZ-VMF3 and TUW-VMF3, the difference in their total arc lengths is not significant, meaning that the RMSE values of the two products are not very different throughout the country. Specifically for each region, the TUW-VMF3 is better than the GFZ-VMF3 in every region, but none differed by more than 2 mm, and their minimum and maximum values appeared both in the TC and SM region. Compared to the TUW-VMF3-$5.0°\times5.0°$, the TUW-VMF3-$1.0°\times1.0°$ has a significantly lower RMSE at the national level, with a shorter total chord length in Fig. 5. Specifically for each region, the RMSE of TUW-VMF3-$1.0°\times1.0°$ is lower than $5.0°\times5.0°$ in each region. Particularly in the SM and TC regions, there are 38.8 % and 46.9 % RMSE decreases from Table.2, respectively. The MP region, which has the smallest change, decreases by 15 %.

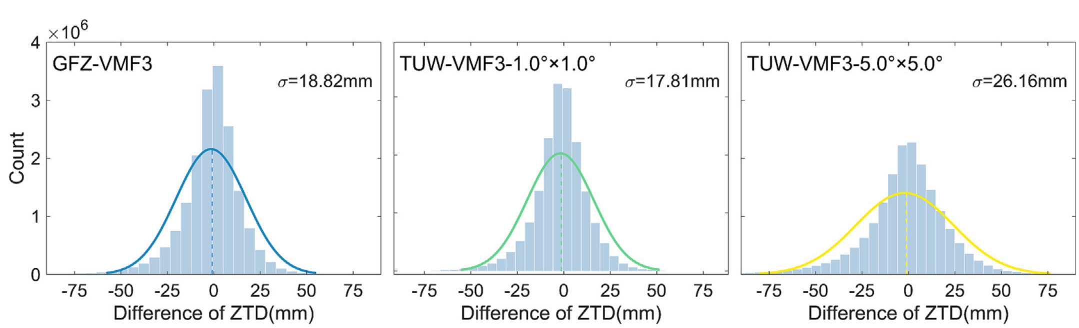

Fig. 3. Histogram of the ZTD difference distribution for the three products.

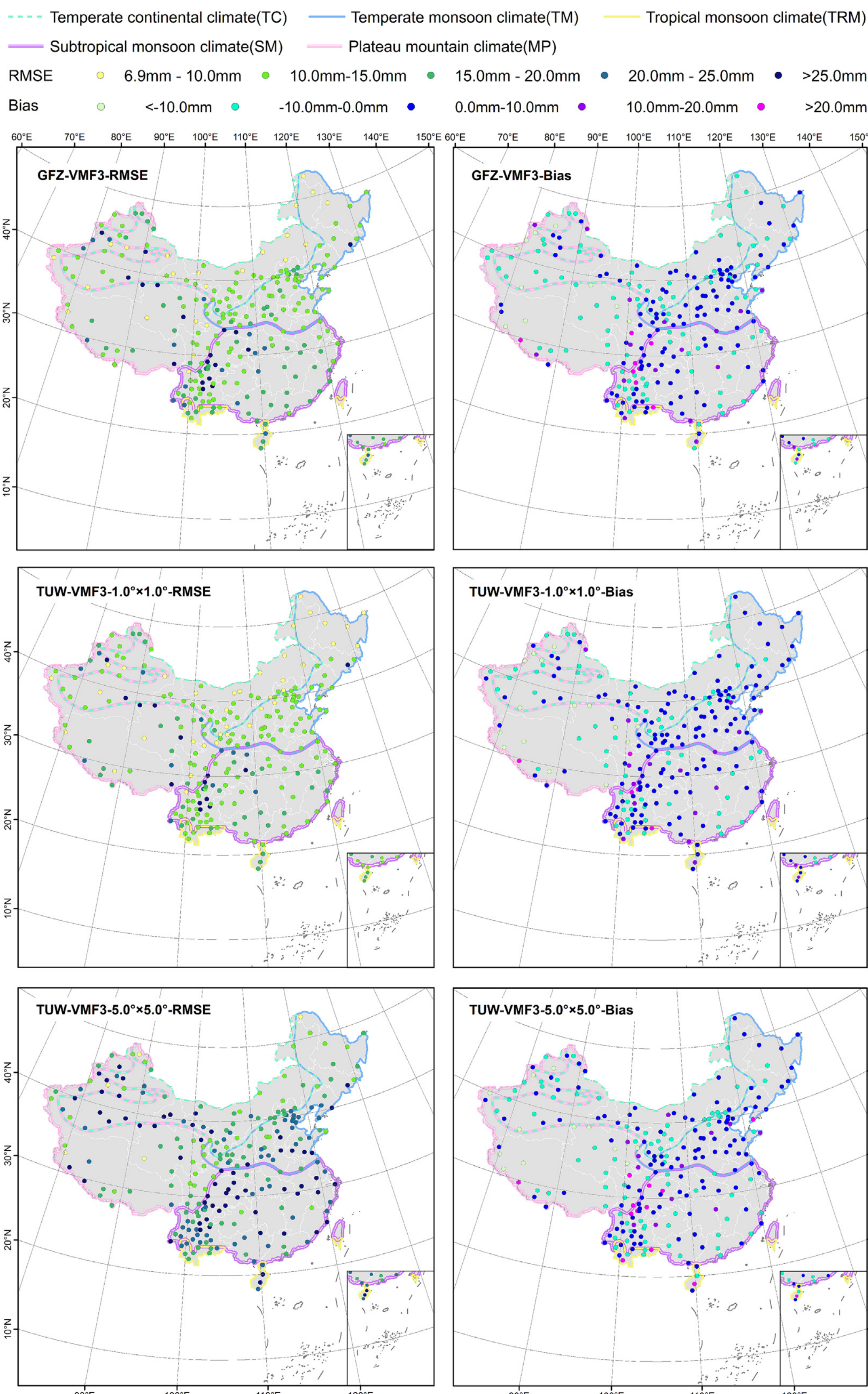

Fig. 4. The distribution of RMSE and bias of the three products in different climate zones.

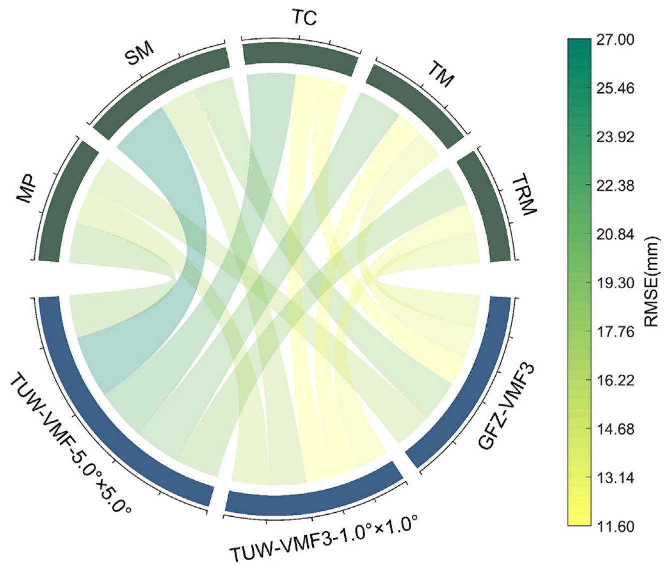

Fig. 5. Descriptions of the RMSE metrics for the three products in different climate zones.

Table 2
Mean RMSE data for the three products in different climatic zones.

Area	GFZ-VMF3 (mm)	TUW-VMF3 −1.0°×1.0° (mm)	TUW-VMF3 −5.0°×5.0° (mm)
MP	16.35	15.51	18.31
SM	17.91	16.48	26.94
TC	12.25	11.64	21.91
TM	13.20	12.07	20.07
TRM	14.20	12.90	18.32

3.3. Spatio-temporal analysis

Fig. 6 illustrates the variation of absolute bias values with altitude, while Table. 3 provides a statistical count of the number of stations within each altitude range. Each spoke represents a distinct range of altitude, and each concentric circle represents a different absolute bias value, increasing in size from zero at the center to the outer circles. As depicted in the figure, the absolute bias of the three products generally decreases with increasing altitude. However, there are notable distinctions in their performance. The absolute bias of GFZ-VMF3 is the highest observed in the 0–100 m altitude range, exceeding 20 mm, while that of TUW-VMF3 is relatively lower, with a maximum value of less than 15 mm. The TUW-VMF3 exhibits the highest absolute bias at 200 m, yet remains below 20 mm. Between 500 and 1000 m, the GFZ-VMF3 and TUW-VMF3 models exhibit variable biases without a discernible trend. Above 1000 m, the GFZ-VMF3 model displays a more significant and consistent reduction in absolute bias compared to the TUW-VMF3 model. Upon further comparison of products with varying resolutions, the absolute bias between the TUW-VMF3-1.0°×1.0° and TUW-VMF3-5.0°×5.0° models is similar, fluctuating within a range of 0–20 mm and peaking at around 200 m. In the altitude range of 0–900 m, the TUW-VMF3-1.0°×1.0° demonstrates a more significant reduction in absolute bias. Conversely, above 900 m, the TUW-VMF3-5.0°×5.0° exhibits a more pronounced decline. This indicates that higher resolution models show greater consistency at lower altitudes, whereas lower resolution models exhibit more regularity at higher altitudes.

To investigate the influence of latitude on the precision of tropospheric delay estimates, Fig. 7 utilizes a dual-axis chart to depict the latitudinal trends in RMSE and bias. The study stratifies latitude into three tiers: below 30°N (A1), between 30°N and 38°N(A2), and above 38°N(A3), with respective station counts of 76, 80, and 82. The left axis denotes the RMSE averaged over nine years for each station, while the right axis signifies the average bias of the qualifying stations. The research indicates no significant variation in RMSE values between the GFZ-VMF3 and TUW-VMF3, with the latter showing a slight reduction in RMSE. Both products exhibit similar trends in RMSE with respect to latitude, generally maintaining values below 30 mm, peaking at around 50 mm, and showing a decrease with increasing latitude. In the comparative analysis, the TUW-VMF3-5.0°×5.0° manifests higher RMSE values, with less distinct patterns, and a significant number of outliers that exceed 60 mm, while the typical values are generally within 40 mm. This observation sug-

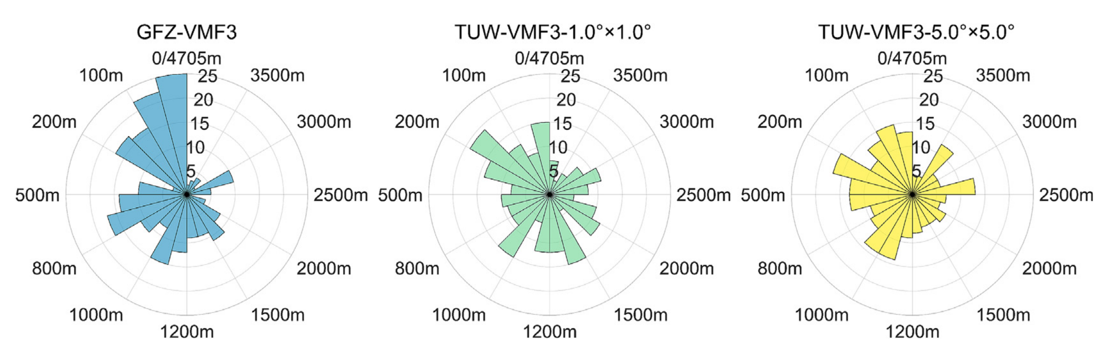

Fig. 6. Absolute bias in mm for three tropospheric products at different altitudes.

Table 3
Station counts by altitude range.

Altitude(m)	0 ~ 100	100 ~ 200	200 ~ 500	500 ~ 800	800 ~ 1000	1000 ~ 1200
StationCounts	47	33	13	31	19	27
Altitude(m)	1200 ~ 1500	1500 ~ 2000	2000 ~ 2500	2500 ~ 3000	3000 ~ 3500	3500 ~ 4705
StationCounts	18	19	5	15	6	5

gests a relative enhancement of approximately 25.0 % in the overall accuracy of the TUW-VMF3-1.0°×1.0° model. Concerning the average bias, all products exhibit a decreasing trend in line with increasing latitude. The GFZ-VMF3 demonstrates a notable increase in bias at lower latitudes, with values exceeding 5 mm below 30°, reducing to below 3 mm between 30° and 38°, and further decreasing to under 1 mm above 38°, suggesting a significant decrease in bias with increasing latitude. The TUW-VMF3 exhibits the average bias is less than 4 mm below 30° latitude, while averaging around 2 mm between 30° and 38° latitudes, and above 38° latitude, indicating a more pronounced latitude-bias correlation at lower latitudes. When comparing products with different spatial resolutions, similar trends in average bias variation with latitude are observed. Notably, at latitudes below 30°, both products show average biases from 3 to 4 mm, which exhibit a decreasing trend with increasing latitude. Within the 30° to 38° and above 38° ranges, biases exhibit minimal variation, with the TUW-VMF3-1.0°×1.0° demonstrating roughly a 33.3 % reduction in bias compared to the TUW-VMF3-5.0°×5.0° across these latitudes, suggesting enhanced precision. It is also noteworthy that all products demonstrate a consistent positive bias, indicating that three products have more tendency to overestimate the ZTD value.

To gain a comprehensive understanding of the annual variations in ZTD, while also considering the impact of errors and data integrity on the results, the two stations, namely the SCSM station with the largest RMSE and XJBL station with the smallest RMSE, were selected to show their ZTD time series from 2012 to 2020 in Fig. 8. The trends and fluctuations in ZTD values for different products at the XJBL and SCSM stations are clearly observable in the figure. Overall, the ZTD values at both stations show an increasing trend at the beginning of the year, reaching a peak around the middle of the year, and

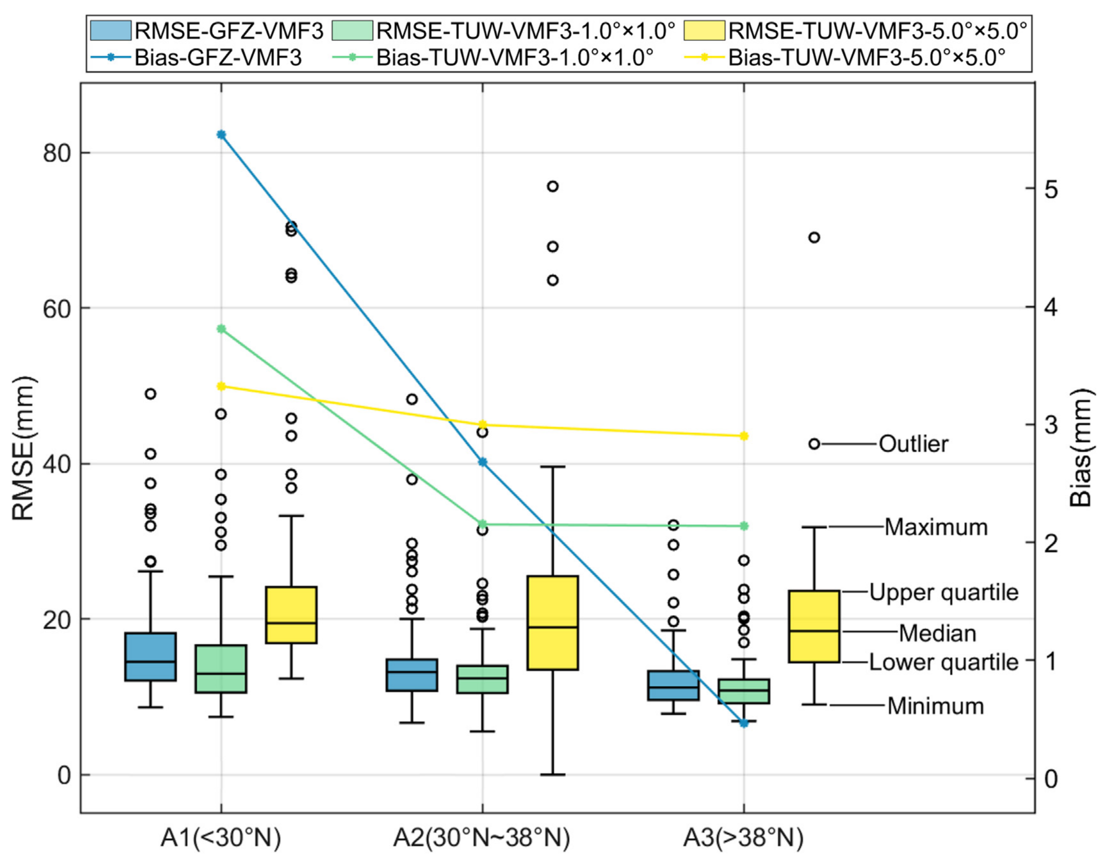

Fig. 7. Variation of precision with latitude for three products.

then decreasing towards the end of the year. Both XJBL and SCSM stations exhibit the largest fluctuations in the middle of the year, but the range of fluctuation at SCSM reaches up to 200 mm, while at XJBL is less than 100 mm. At the SCSM station, the ZTD values of GFZ-VMF3 are generally slightly higher than those of TUW-VMF3, and both are generally higher than the reference ZTD, with differences within 100 mm. There is not much difference between products of different spatial resolutions, but the TUW-VMF3-5.0°×5.0° often has higher ZTD values in the middle of the year and lower ZTD values at the beginning and end of the year. At the XJBL station, the ZTD values of GFZ-VMF3 and TUW-VMF3 are highly consistent and closely match the reference ZTD. Compared to the TUW-VMF3-1.0°×1.0°, the ZTD values of TUW-VMF3-5.0°×5.0° are generally lower. Residual analysis indicates that products with the same spatial resolution exhibit a considerable overlap in results. Among products of different spatial resolutions, the product with higher spatial resolution demonstrates more concentrated residuals. At the SCSM station, the most residual values are concentrated between −100 mm and 100 mm, but some residual values in the middle of the year are between −100 mm and −200 mm. At the XJBL station, the residual values are more stable, mainly distributed between −50 mm and 50 mm, and close to 0 mm at the beginning and end of the year. These results indicate that the ZTD values and the residual variations at both stations are closely related to the season, with the ZTD values at the XJBL station being more stable.

Fig. 9 demonstrates the comparative performance of three ZTD products across China. In general, the TUW-VMF3-1.0°×1.0° stands out among the three, achieving optimal performance at 82.4 % of the stations nationwide. Its performance is notably superior in each region, with particularly strong results in the TM and SM regions. When comparing products of the same resolution, the TUW-VMF3 shows a superior performance compared to the GFZ-VMF3 at 89.1 % of the stations. Especially in the SM, TM, and TRM regions, where the TUW-VMF3 almost uniformly exhibits higher precision at all stations, with only one or two exceptions. Compared to GFZ-VMF3, the GFZ-VMF3 has higher precision in the western regions of the TC and MP areas, and 80.8 % of the stations are above 1000 m. Regarding products of different resolutions, the TUW-VMF3-1.0°×1.0° outperforms the TUW-VMF3-5.0°×5.0° at 93.3 % of the stations nationwide, especially in the TC, TM, and SM regions, where almost all stations have seen a significant improvement in precision, with only one or two exceptions. In the TRM region, the performance optimization of the stations even reached 100 %. However, there are still 6.7 % of stations that are not improved, of which 56.3 % are above 1500 m in altitude, and among the lower altitude stations, 42.9 % are located in areas with significant terrain undulations such as Altay. This may be due to the fact that in these loca-

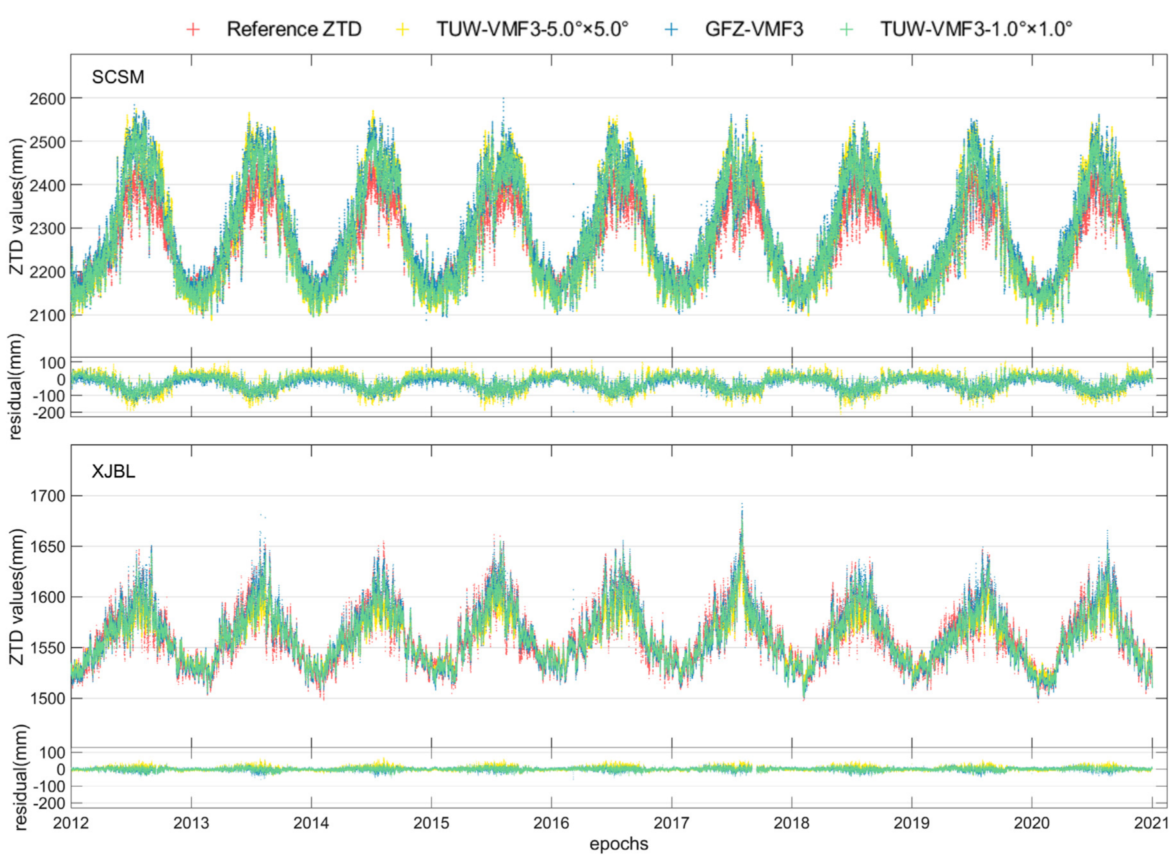

Fig. 8. ZTD time series at SCSM and XJBL stations showing relatively high and low RMSE values.

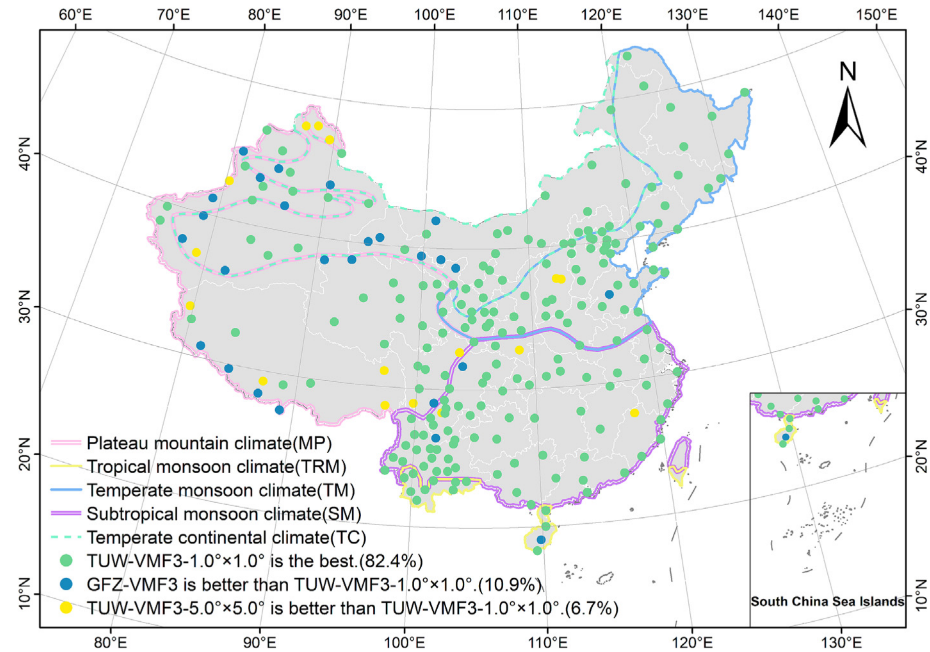

Fig. 9. Analysis of the performance of three ZTD products, focusing on the strengths and weaknesses of the TUW-VMF3-1.0°×1.0°.

tions, both the TUW-VMF3-1.0°×1.0° and the TUW-VMF3-5.0°×5.0° struggle to accurately represent the height difference relationship between the grid and the target, resulting in a loss of precision. It is worth noting that the TUW-VMF3-1.0°×1.0° performs best at all stations below 200 m in altitude. These results emphasize that the advantage of the TUW-VMF3-1.0°×1.0° in low-altitude areas is greater than in high-altitude areas.

Fig. 10 analyzes the accuracy variations of the three products across different seasons. The results reveal similar and pronounced seasonal and regional characteristics in the RMSE metrics of all products. Specifically, the RMSE of these products typically peaks in summer and reaches its minimum in winter, with the SM region consistently exhibiting higher RMSE values throughout the seasons compared to other regions. However, this pattern is less distinct in the TRM region. For GFZ-VMF3 and TUW-VMF3, the seasonal trends in RMSE are consistent across all regions, with minimal variance in the specific indicator values, especially in the TC region, but usually the TUW-VMF3 typically exhibits slightly lower RMSE than the GFZ-VMF3. The RMSE in MP and TRM regions remains stable throughout the seasons with minimal fluctuations. Comparing products of different resolutions, the TUW-VMF3-1.0°×1.0° generally demonstrates lower RMSE values than the TUW-VMF3-5.0°×5.0°, especially in the SM, TC, and TM regions during summer, where differences exceed 10 mm, indicating a significant enhancement in data accuracy for the TUW-VMF3-1.0°×1.0°. It is observed that three products achieve their lowest RMSE in the TM region during winter, while the RMSE in the TRM region is unusually high in spring and autumn, potentially due to the ambiguous seasonal boundaries in this area.

4. Conclusion

In order to evaluate the accuracy of the ZTD values of the TUW-VMF3 and GFZ-VMF3 products in China region, this paper selects the hourly ZTD values as the estimated ZTD values and makes a comprehensive comparison with the ZTD values of 238 GNSS stations in the existing ZTD data of CMONOC for the period of 2012–

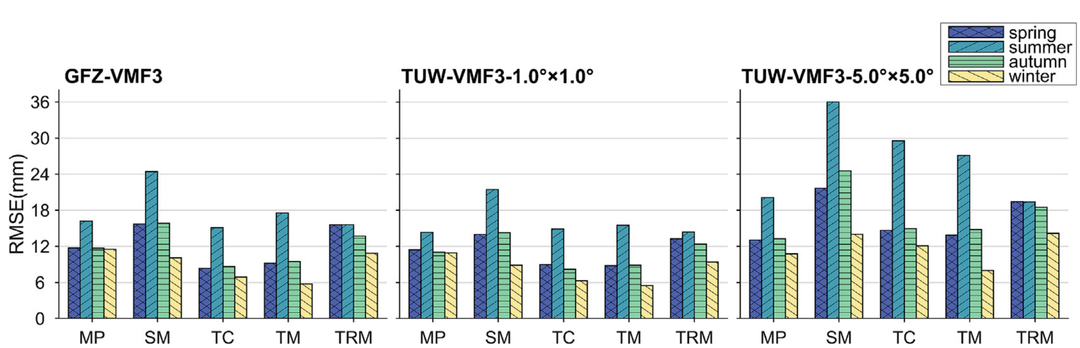

Fig. 10. RMSE for three tropospheric products in different seasons and regions.

2020 as a reference. In this process, particular attention is given to products with varying spatial resolutions to ensure the comprehensiveness and accuracy of the assessment results.

For products of the same spatial resolution, the average RMSE/Bias of the TUW-VMF3/GFZ-VMF3 products was 17.90/1.82 and 18.87/1.42 mm, respectively. The differences between two products and the reference ZTD values show a normal distribution, which accounts for over 98 % ranging from −50 mm to 50 mm. The performance of both products was similarly distributed across China. In terms of RMSE, they performed well in the central and northeastern regions of China, where RMSE values were below 15 mm. Regarding bias, the values were concentrated between −10 and 10 mm, with a tendency to underestimate in the northwest and overestimate in the central and northeastern regions, while the southeast exhibited varying biases. Among all GNSS stations, TUW-VMF3 outperformed GFZ-VMF3 at 89.1 % of the stations. For products of different spatial resolutions, the TUW-VMF3-5.0°×5.0° had an overall RMSE that was 8.34 mm higher than the TUW-VMF3-1.0°×1.0°, with a negligible bias difference of 0.18 mm. Moreover, the TUW-VMF3-1.0°×1.0° significantly improved accuracy at 93.3 % of the stations compared to the TUW-VMF3-5.0°×5.0°. The remaining 6.7 % of stations, primarily located in extreme high-altitude or extremely cold regions, showed less improvement with the TUW-VMF3-1.0°×1.0°. These stations were categorized into five regions: TRM, SM, TM, TC, and MP. For products of the same resolution, both achieved good accuracy in the TC and TM regions, but TUW-VMF3 outperformed GFZ-VMF3 in each region, with RMSE differences not exceeding 2 mm. For products of different resolutions, the TUW-VMF3-1.0°×1.0° significantly outperformed the TUW-VMF3-5.0°×5.0°, especially in the SM and TC regions, where the RMSE was reduced by 38.8 % and 46.9 %, respectively.

Considering seasonal factors, all regions performed worst in summer and best in winter, with the exception of the TRM region. Additionally, for products of the same spatial resolution, except for the SM region, GFZ-VMF3 and TUW-VMF3 performed very well in other regions, particularly in the TC and TM regions during winter. For products of different spatial resolutions, the TUW-VMF3-1.0°×1.0° showed significant improvement in all seasons, especially in the SM, TC, and TM regions during summer. Furthermore, the RMSE values of all three products decreased with increasing altitude and latitude, with the most pronounced change observed in GFZ-VMF3. Time series analysis of ZTD at stations with extreme RMSE values indicated that ZTD values were highest in summer and lowest in winter.

These findings imply that the TUW-VMF3-1.0°×1.0° is recommended for priority use due to that it outperforms the other products in most scenarios. Meanwhile, it is suggested to fully consider the characteristics of different climate regions and seasonal factors to comprehensively select the most suitable products for improving ZTD estimation in GNSS applications in China. Due to the limited number of CMONOC stations, the coverage of this study is limited, and the time span of the study still has room for improvement. With the continuous accumulation and enrichment of GNSS data, it is expected that a more in-depth and comprehensive comparative analysis of the three products can be conducted over a wider area and a longer time scale in the future.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors would like to thank the CMONOC, TUW, and GFZ for providing the study dataset. This research was founded by the National Natural Science Foundation of China (42204022), Science and Technology Development Plan Project of the Silk Road Economic Belt Innovation-Driven Development Zone and WuChangShi National Independent Innovation Demonstration Zone (2023LQY02), the Open Research Fund of State Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University (23P02), the Fundamental Research Funds for the Central Universities (2024ZKPYDC02, 2024CXCYY010, 2042023kf0003), China University of Mining and Technology-Beijing Innovation Training Program for College Students (202402008, 202402010).

References

Bevis, M., Businger, S., Herring, T.A., Rocken, C., Anthes, R.A., Ware, R.H., 1992. GPS meteorology: remote sensing of atmospheric water vapor using the global positioning system. J. Geophys. Res.: Atmosph. 97 (D14), 15787–15801. https://doi.org/10.1029/92JD01517.

Black, H.D., 1978. An easily implemented algorithm for the tropospheric range correction. J. Geophys. Res. Atmos. 83 (B4), 1825–1828. https://doi.org/10.1029/JB083iB04p01825.

Boehm, J., Kouba, J., Schuh, H., 2009. Forecast Vienna Mapping Functions 1 for real-time analysis of space geodetic observations. J. Geod 83, 397–401. https://doi.org/10.1007/s00190-008-0216-y.

Böhm, J., Möller, G., Schindelegger, M., Pain, G., Weber, R., 2015. Development of an improved empirical model for slant delays in the troposphere (GPT2w). GPS Solut. 19 (3), 433–441. https://doi.org/10.1007/s10291-014-0403-7.

Chen, P., Ma, Y., Liu, H., Zheng, N., 2020. A new global tropospheric delay model considering the spatiotemporal variation characteristics of ZTD with altitude coefficient. Earth Space Sci. 7 (4). https://doi.org/10.1029/2019EA000888 e2019EA000888.

Chen, J., Wang, J., Wang, J., Tan, W., 2019. SHAtrop: Empirical ZTD Model Based on CMONOC GNSS Network. Geomat. Inform. Sci. Wuhan Univ. 44 (11), 1588–1595. https://doi.org/10.13203/j.whugis20170384.

Geng, J., Chen, X., Pan, Y., Mao, S., Li, C., Zhou, J., Zhang, K., 2019. PRIDE PPP-AR: An open-source software for GPS PPP ambiguity

resolution. GPS Solut. 23, 91. https://doi.org/10.1007/s10291-019-0888-1.
Glaner, M.F., Weber, R., 2023. An open-source software package for precise point positioning: RaPPPid. GPS Solut. 27 (4), 174. https://doi.org/10.1007/s10291-023-01488-4.
He, L., Yao, Y., Xu, C., Zhang, H., Tang, F., Ji, C., Liu, Z., Wu, W., 2024. A new global ZTD forecast model based on improved LSTM neural network. IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens. PP (99), 1–10. https://doi.org/10.1109/JSTARS.2024.3391821.
Hopfield, H.S., 1969. Two-quartic tropospheric refractivity profile for correcting satellite data. J. Geophys. Res. Atmos. 74 (18), 4487–4499. https://doi.org/10.1029/JC074i018p04487.
Jiang, C., Xu, T., Wang, S., Nie, W., Sun, Z., 2020. Evaluation of zenith tropospheric delay derived from ERA5 data over china using GNSS observations. Remote Sens. (Basel) 12 (4), 663. https://doi.org/10.3390/rs12040663.
Lagler, K., Schindelegger, M., Böhm, J., Krásná, H., Nilsson, T., 2013. GPT2: Empirical slant delay model for radio space geodetic techniques. Geophys. Res. Lett. 40 (6), 1069–1073. https://doi.org/10.1002/grl.50288.
Leandro, R.F., Langley, R.B., Santos, M.C., 2008. UNB3m_pack: a neutral atmosphere delay package for radiometric space techniques. GPS Solutions 12, 65–70. https://doi.org/10.1007/s10291-007-0077-5.
Leandro, R., Santos, M., & Langley, R. B. 2006. UNB Neutral Atmosphere Models: Development and Performance.
Li, J., Yang, F., Yuan, D., Wang, H., Song, S., Tan, J., Wen, Z., 2024. Unraveling the accuracy enigma: investigating ZTD data precision in TUW-VMF3 and GFZ-VMF3 products using a comprehensive global GPS dataset. IEEE Trans. Geosci. Remote Sens. PP (99), 1. https://doi.org/10.1109/TGRS.2024.3385228.
Li, W., Yuan, Y., Ou, J., Li, H., Li, Z., 2012. A new global zenith tropospheric delay model IGGtrop for GNSS applications. Chin. Sci. Bull. 57, 2132–2139. https://doi.org/10.1007/s11434-012-5010-9.
Osah, S., Acheampong, A.A., Fosu, C., Dadzie, I., 2021. Evaluation of zenith tropospheric delay derived from ray-traced VMF3 product over the west african region using GNSS observations. Adv. Meteorol. 2021 (6), 1–14. https://doi.org/10.1155/2021/8836806.
Penna, N., Dodson, A., Chen, W., 2001. Assessment of EGNOS tropospheric correction model. J. Navig. 54 (1), 37–55. https://doi.org/10.1017/S0373463300001107.
Saastamoinen, J. H. 1972. Atmospheric Correction for the Troposphere and the Stratosphere in Radio Ranging Satellites. doi:10.1029/GM015p0247.
Song, S., Zhu, W., Chen, Q., Liou, Y.A., 2011. Establishment of a new tropospheric delay correction model over China area. Science China Physics Mechanics & Astronomy 54 (12). https://doi.org/10.1007/s11433-011-4530-7.
Ssenyunzi, R.C., Andima, G., Amabayo, E., Realini, E., 2023. Performance of ray-traced VMF3 products in retrieving Zenith Tropospheric Delay over the African tropical region. J. Atmos. Sol. Terr. Phys. 243 (2). https://doi.org/10.1016/j.jastp.2023.106014 106014.
Wang, Y., Yang, F., Li, P., Gong, X., Liu, M., Xu, T., Lin, X., Wang, Y., 2024. An optimal calibration method for MODIS precipitable water vapor using GNSS observations. Atmos. Res. 309. https://doi.org/10.1016/j.atmosres.2024.107591 107591.
Wang, X., Zhu, G., Huang, L., Wang, H., Yang, Y., Li, J., Huang, L., Zhou, L., Liu, L., 2022. Development of a ZTD vertical profile model considering the spatiotemporal variation of height scale factor with different reanalysis products in China. Atmos. 13 (9), 1469. https://doi.org/10.3390/atmos13091469.
Yang, L., Fu, Y., Zhu, J., Shen, Y., Rizos, C., 2023b. Overbounding residual zenith tropospheric delays to enhance GNSS integrity monitoring. GPS Solutions 27 (2). https://doi.org/10.1007/s10291-023-01408-6.
Yang, F., Guo, J., Zhang, C., Li, Y., Li, J., 2021a. A Regional Zenith Tropospheric Delay (ZTD) Model Based on GPT3 and ANN. Remote Sens. 13 (5), 838. https://doi.org/10.3390/rs13050838.
Yang, F., Meng, X., Guo, J., Yuan, D., Chen, M., 2021b. Development and evaluation of the refined zenith tropospheric delay (ZTD) models. Satell. Navig. 2 (1), 21. https://doi.org/10.1186/s43020-021-00052-0.
Yang, F., Sun, Y., Meng, X., Guo, J., Gong, X., 2023a. Assessment of tomographic window and sampling rate effects on GNSS water vapor tomography. Satell. Navig. 4, 7. https://doi.org/10.1186/s43020-023-00096-4.
Yang, F., Liu, M., Zhao, Y., An, X., Wang, L., Wen, Z., 2024. Higher accuracy estimation of the weighted mean temperature (Tm) using GPT3 model with new grid coefficients over China. Atmos. Res. 305. https://doi.org/10.1016/j.atmosres.2024.107424 107424.
Yang, L., Wang, J., Li, H., Balz, T., 2021c. Global assessment of the GNSS single point positioning biases produced by the residual tropospheric delay. Remote Sens. (Basel) 13 (6), 1202. https://doi.org/10.3390/rs13061202.
Yao, Y., He, C., Zhang, B., Xu, C., 2013. A new global zenith tropospheric delay model GZTD. Chinese Journal of Geophysics (in Chinese) 56 (7), 2218–2227. https://doi.org/10.6038/cjg20130709.
Yao, Y., Sun, Z., Xu, C.Q., 2018. Establishment and evaluation of a new meteorological observation-based grid model for estimating zenith wet delay in ground-based global navigation satellite system (GNSS). Remote Sens. (Basel) 10 (11), 1718. https://doi.org/10.3390/rs10111718.
Yao, Y., Yang, F., Li, J., Wang, L., Zheng, J., Hao, R., Xu, T., 2024. Comparing discrete and empirical troposphere delay models: a global IGS-based evaluation. Radio Sci. 59. https://doi.org/10.1029/2024RS007950.
Zhang, H., Yuan, Y., Li, W., 2021. An analysis of multisource tropospheric hydrostatic delays and their implications for GPS/GLONASS PPP-based zenith tropospheric delay and height estimations. J. Geod. 95 (7), 83. https://doi.org/10.1007/s00190-021-01535-3.
Zumberge, J.F., Heflin, M.B., Jefferson, D.C., Watkins, M.M., Webb, F. H., 1997. Precise point positioning for the efficient and robust analysis of GPS data from large networks. J. Geophys. Res. Atmos. 102 (B3), 5005–5017. https://doi.org/10.1029/96JB03860.
Zus, F., Dick, G., Dousa, J., Wickert, J., 2015. Systematic errors of mapping functions which are based on the VMF1 concept. GPS Solutions 19 (2), 277–286. https://doi.org/10.1007/s10291-014-0386-4.